

MITRE ATT&CK Threat Detection Report

Environment: WIN11-TARGET (Windows 11 Pro VM)

Tools Used: Splunk Enterprise 10.4, Sysmon v15 (Olaf Hartong config)

Date: June 2026

Author: Michael Isiuwa

Executive Summary

This report documents the simulation and detection of a three-stage attack chain using the MITRE ATT&CK framework. The objective was to emulate realistic adversary behaviour across three distinct tactics - Execution, Persistence, and Defense Evasion and demonstrate detection capability using Splunk Enterprise and Sysmon on a controlled Windows 11 lab environment.

Each technique was manually simulated, captured in Sysmon logs, detected via targeted Splunk queries, and mapped to its corresponding MITRE ATT&CK technique ID. Detection rules were created in Splunk to provide ongoing alerting capability beyond the initial simulation.

The attack chain simulated represents a realistic post-compromise scenario: an attacker executes malicious PowerShell commands after gaining initial access, establishes persistence via a scheduled task, then attempts to evade detection by masquerading a malicious executable as a legitimate Windows process.

Attack Chain Overview

Stage Technique		MITRE ID Tactic	
1	PowerShell Execution	T1059.001	Execution
2	Scheduled Task Creation	T1053.005	Persistence
3	Masquerading	T1036.005	Defense Evasion

Environment & Tools

Lab Architecture

The simulation was conducted in an isolated virtual machine environment using Oracle VirtualBox. No external networks or production systems were involved. All attack activity was contained within the virtual machine.

Component	Details
Host OS	Windows 11 (main PC)
Virtual Machine	Windows 11 Pro — WIN11-TARGET
Hypervisor	Oracle VirtualBox 7.x
RAM Allocated	6 GB
Disk Allocated	80 GB

Security Tools

Sysmon v15 (System Monitor)

Microsoft Sysinternals tool deployed on WIN11-TARGET to provide enhanced Windows event logging. Configured using the Olaf Hartong sysmon-modular configuration — the most widely adopted community Sysmon config, tuned to detect MITRE ATT&CK techniques and automatically tag events with technique IDs. Sysmon captured process creation, file creation, registry modification, and network events throughout the simulation.

Splunk Enterprise 10.4

Industry-leading SIEM platform used to ingest, search, and alert on Sysmon logs. Configured to monitor the Microsoft-Windows-Sysmon/Operational event log channel. All detections were performed using SPL (Search Processing Language) queries. Three scheduled alert rules were created — one per technique — to provide ongoing detection capability.

Technique 1: PowerShell Execution — T1059.001

Field	Details
Technique	PowerShell Execution
MITRE ID	T1059.001
Tactic	Execution
Severity	High

Description

PowerShell is a powerful scripting environment built into every Windows system. Attackers abuse it to execute malicious commands, download payloads, and interact with the operating system while blending in with legitimate administrative activity. The use of the -ExecutionPolicy Bypass flag is a well-known attacker technique to override Windows script execution restrictions.

Attack Simulation

The following PowerShell command was executed on WIN11-TARGET to simulate post-compromise reconnaissance activity:

```
powershell -ExecutionPolicy Bypass -NoProfile -Command "Get-Process; Get-LocalUser; whoami"
```

This command specifies running processes, lists local user accounts, and confirms the current user context - standard attacker behaviour immediately after gaining access to a system.

Detection

Sysmon EventID 1 (Process Creation) captured the execution and automatically mapped it to MITRE T1059.001. The following Splunk query was used to detect the activity:

```
index=main source="WinEventLog:Microsoft-Windows-Sysmon/Operational"  
"ExecutionPolicy" "Bypass"
```

Splunk returned 6 events confirming detection of the simulated attack. Key fields observed in the logs:

Field	Value
Image	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
CommandLine	powershell.exe -ExecutionPolicy Bypass -NoProfile
ParentImage	C:\Windows\System32\cmd.exe
User	WIN11-TARGET\analyst

Field	Value
MITRE Tag	technique_id=T1059.001, technique_name=PowerShell

i	Time	Event
>	6/16/26 3:03:41.906 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5778385f-c22a-43e0-bf4c-06f5698fbd9}'><EventID>1</EventID><Version>5</Version><Level>4</Level><Task>1</Task><OpCode>0</OpCode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-06-16T19:03:41.9068715Z'><EventRecordID>27351</EventRecordID><Correlation><Execution ProcessID='1256' ThreadID='4036'><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>WIN11-TARGET</Computer><Security UserID='S-1-5-18'></System><EventData><Data Name='RuleName'>technique_id=T1033,technique_name=System Owner/User Discovery</Data><Data Name='UtcTime'>2026-06-16 19:03:41.900</Data><Data Name='ProcessGuid'>{878543f7-9e0d-6a31-931f-000000000000}</Data><Data Name='ProcessId'>5384</Data><Data Name='Image'>C:\Windows\System32\whoami.exe</Data><Data Name='FileVersion'>10.0.26100.1882 (WinBuild.160101.0800)</Data><Data Name='Description'>whoami - displays logged on user information</Data><Data Name='Product'>Microsoft® Windows® Operating System</Data><Data Name='Company'>Microsoft Corporation</Data><Data Name='OriginalFileName'>whoami.exe</Data><Data Name='CommandLine'>"C:\WINDOWS\system32\whoami.exe"</Data><Data Name='CurrentDirectory'>C:\WINDOWS\system32</Data><Data Name='User'>WIN11-TARGET\analyst</Data><Data Name='LogonId'>{878543f7-8c51-6a31-195c-0e0000000000}</Data><Data Name='LogonId'>0xe5c19</Data><Data Name='TerminalSessionId'>1</Data><Data Name='IntegrityLevel'>High</Data><Data Name='Hashes'>SHA1=C23488BA47972B04F795D34B35C8257EC4B7ACD0_MD5=956692DADC582CEB46E9219F7A5BEFFA_SHA256=23248EF9F880A9A324110B1C2331DE31DC180E08F5359CB707E51A939AF56CD3_IMPHASH=E4464BCD920F7AC69CCD874CB9C0FEF</Data><Data Name='ParentProcessGuid'>{878543f7-9e09-6a31-901f-000000000000}</Data><Data Name='ParentProcessId'>6580</Data><Data Name='ParentImage'>C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe</Data><Data Name='ParentCommandLine'>"C:\WINDOWS\system32\WindowsPowerShell\v1.0\powershell.exe" -ExecutionPolicy Bypass -NoProfile -Command "Get-Process; Get-LocalUser; whoami"</Data><Data Name='ParentUser'>WIN11-TARGET\analyst</Data></EventData></Event> host = WIN11-TARGET source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational
>	6/16/26 3:03:37.977 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5778385f-c22a-43e0-bf4c-06f5698fbd9}'><EventID>1</EventID><Version>5</Version><Level>4</Level><Task>1</Task><OpCode>0</OpCode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-06-16T19:03:37.9775400Z'><EventRecordID>27335</EventRecordID><Correlation><Execution ProcessID='1256' ThreadID='4036'><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>WIN11-TARGET</Computer><Security UserID='S-1-5-18'></System><EventData><Data Name='RuleName'>technique_id=T1059.001,technique_name=PowerShell</Data><Data Name='UtcTime'>2026-06-16 19:03:37.969</Data><Data Name='ProcessGuid'>{878543f7-9e09-6a31-901f-000000000000}</Data><Data Name='ProcessId'>6580</Data><Data Name='Image'>C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe</Data><Data Name='FileVersion'>10.0.26100.5074 (WinBuild.160101.0800)</Data><Data Name='Description'>Windows PowerShell</Data><Data Name='Product'>Microsoft® Windows® Operating System</Data><Data Name='Company'>Microsoft Corporation</Data><Data Name='OriginalFileName'>PowerShell.EXE</Data><Data Name='CommandLine'>"C:\WINDOWS\system32\WindowsPowerShell\v1.0\powershell.exe" -ExecutionPolicy Bypass -NoProfile -Command "Get-Process; Get-LocalUser; whoami"</Data><Data Name='CurrentDirectory'>C:\WINDOWS\system32</Data><Data Name='User'>WIN11-TARGET\analyst</Data><Data Name='LogonId'>{878543f7-8c51-6a31-195c-0e0000000000}</Data><Data Name='LogonId'>0xe5c19</Data><Data Name='TerminalSessionId'>1</Data><Data Name='IntegrityLevel'>High</Data><Data Name='Hashes'>SHA1=EB42621654E82FAF20E340442B60EB1A77664E5B_MD5=A97E657B97B44C96122BF4543AB2CA1_SHA256=0FF62C94BC7E283A5F7E160E1622E50BA70396F31C7D5F56381870317E8C46_IMPHASH=AFAC6DC9A4114B198160AB4D0AE77</Data><Data Name='ParentProcessGuid'>{878543f7-99f5-6a31-1910-000000000000}</Data><Data Name='ParentProcessId'>6020</Data><Data Name='ParentImage'>C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe</Data><Data Name='ParentCommandLine'>"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"</Data><Data Name='ParentUser'>WIN11-TARGET\analyst</Data></EventData></Event> host = WIN11-TARGET source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational
>	6/16/26 3:03:25.221 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5778385f-c22a-43e0-bf4c-06f5698fbd9}'><EventID>1</EventID><Version>5</Version><Level>4</Level><Task>1</Task><OpCode>0</OpCode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-06-16T19:03:25.2217627Z'><EventRecordID>27328</EventRecordID><Correlation><Execution ProcessID='1256' ThreadID='4036'><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>WIN11-TARGET</Computer><Security UserID='S-1-5-18'></System><EventData><Data Name='RuleName'>technique_id=T1033,technique_name=System Owner/User Discovery</Data><Data Name='UtcTime'>2026-06-16 19:03:25.213</Data><Data Name='ProcessGuid'>{878543f7-90fd-6a31-751f-000000000000}</Data><Data Name='ProcessId'>724</Data><Data Name='Image'>C:\Windows\System32\whoami.exe</Data><Data Name='FileVersion'>10.0.26100.1882 (WinBuild.160101.0800)</Data><Data Name='Description'>whoami - displays logged on user information</Data><Data Name='Product'>Microsoft® Windows® Operating System</Data><Data Name='Company'>Microsoft Corporation</Data><Data Name='OriginalFileName'>whoami.exe</Data><Data Name='CommandLine'>"C:\WINDOWS\system32\whoami.exe"</Data><Data Name='CurrentDirectory'>C:\WINDOWS\system32</Data><Data Name='User'>WIN11-TARGET\analyst</Data><Data Name='LogonId'>{878543f7-8c51-6a31-195c-0e0000000000}</Data><Data Name='LogonId'>0xe5c19</Data><Data Name='TerminalSessionId'>1</Data><Data Name='IntegrityLevel'>High</Data><Data Name='Hashes'>SHA1=EB42621654E82FAF20E340442B60EB1A77664E5B_MD5=A97E657B97B44C96122BF4543AB2CA1_SHA256=0FF62C94BC7E283A5F7E160E1622E50BA70396F31C7D5F56381870317E8C46_IMPHASH=AFAC6DC9A4114B198160AB4D0AE77</Data><Data Name='ParentProcessGuid'>{878543f7-99f5-6a31-1910-000000000000}</Data><Data Name='ParentProcessId'>6020</Data><Data Name='ParentImage'>C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe</Data><Data Name='ParentCommandLine'>"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"</Data><Data Name='ParentUser'>WIN11-TARGET\analyst</Data></EventData></Event> host = WIN11-TARGET source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational

Figure 1: Splunk detection of T1059.001 — 6 events returned showing ExecutionPolicy Bypass in CommandLine field

Searches, Reports, and Alerts									
Searches, reports, and alerts are saved searches created from pivot or the search page. Learn more									
1 Searches, Reports, and Alerts Type: All App: Search & Reporting (search) Owner: Administrator (admin) T1059 10 per page									
Name	Actions	Type	Next Scheduled Time	Display View	Owner	App	Alerts	Sharing	Status
Suspicious PowerShell Execution - T1059.001 Detects PowerShell execution with ExecutionPolicy Bypass flag. MITRE ATTACK T1059.001	Edit Run View Recent	Alert	2026-06-23 16:00 Eastern Daylight Time	none	admin	search	0	Private	Enabled

Alert screenshot: Figure 2: Splunk alert rule "Suspicious PowerShell Execution - T1059.001" configured to run hourly

Detection Rule

A scheduled Splunk alert was created titled "Suspicious PowerShell Execution - T1059.001" configured to run hourly and trigger when results exceed zero.

Recommended Mitigation

- Restrict PowerShell execution policy via Group Policy to AllSigned or RemoteSigned
- Enable PowerShell Script Block Logging (Event ID 4104) for deeper visibility
- Alert on any use of -ExecutionPolicy Bypass in production environments

- Deploy Constrained Language Mode to limit PowerShell capabilities for standard users

Technique 2: Scheduled Task Creation — T1053.005

Field	Details
Technique	Scheduled Task/Job
MITRE ID	T1053.005
Tactic	Persistence
Severity	High

Description

Scheduled tasks are a native Windows feature that attackers abuse to maintain persistence on a compromised system. By creating a task that executes malicious code at system startup or user logon, an attacker ensures their access survives reboots and user logoffs. Attackers commonly use task names that mimic legitimate Windows maintenance tasks to avoid detection.

Attack Simulation

The following command was executed to create a persistence mechanism on WIN11-TARGET:

```
schtasks /create /tn "WindowsUpdateCheck" /tr "powershell.exe -WindowStyle Hidden -Command 'whoami'" /sc  
onlogon /ru System
```

The task was named WindowsUpdateCheck to blend in with legitimate Windows update activity. It was configured to run as SYSTEM at every user logon, executing a hidden PowerShell command.

Detection

Sysmon captured the attack across three event types simultaneously:

Sysmon Event	ID	What It Captured
Process Creation	EventID 1	schtasks.exe spawned by PowerShell
Registry Modification	EventID 13	Task written to TaskCache registry key
File Creation	EventID 11	Task file created in C:\Windows\System32\Tasks\

The following Splunk query was used to detect the activity:

```
index=main source="WinEventLog:Microsoft-Windows-Sysmon/Operational" "WindowsUpdateCheck"
```

Key fields observed in EventID 1:

Field	Value
Image	C:\Windows\System32\schtasks.exe
CommandLine	schtasks.exe /create /tn WindowsUpdateCheck
ParentImage	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
User	WIN11-TARGET\analyst
MITRE Tag	technique_id=T1053.005, technique_name=Scheduled Task/Job

i	Time	Event
>	6/22/26 2:06:31.407 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698f7bd9}' /><EventID>11</EventID><Version>2</Version><Level>4</Level><Task>13</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-06-22T18:06:31.4078616Z' /><EventRecordID>34932</EventRecordID><Correlation><Execution ProcessID='3176' ThreadID='4132' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>WIN11-TARGET</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='RuleName'>-</Data><Data Name='UtcTime'>2026-06-22 18:06:31.403</Data><Data Name='ProcessGuid'>{878543f7-7907-6a39-1b00-000000000000}</Data><Data Name='ProcessId'>1396</Data><Data Name='Image'>C:\WINDOWS\system32\svchost.exe</Data><Data Name='TargetFileName'>C:\Windows\System32\Tasks\WindowsUpdateCheck</Data><Data Name='CreationUtcTime'>2026-06-22 18:06:31.403</Data><Data Name='User'>NT AUTHORITY\SYSTEM</Data></EventData></Event> host = WIN11-TARGET source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational
>	6/22/26 2:06:31.407 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698f7bd9}' /><EventID>13</EventID><Version>2</Version><Level>4</Level><Task>13</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-06-22T18:06:31.4073724Z' /><EventRecordID>34931</EventRecordID><Correlation><Execution ProcessID='3176' ThreadID='4132' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>WIN11-TARGET</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='RuleName'>technique_id=T1053, technique_name=Scheduled Task</Data><Data Name='EventType'>SetValue</Data><Data Name='UtcTime'>2026-06-22 18:06:31.408</Data><Data Name='ProcessGuid'>{878543f7-7907-6a39-1b00-000000000000}</Data><Data Name='ProcessId'>1396</Data><Data Name='Image'>C:\WINDOWS\system32\svchost.exe</Data><Data Name='TargetObject'>HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tree\WindowsUpdateCheck\index</Data><Data Name='Details'>DWORD (0x00000002)</Data><Data Name='User'>NT AUTHORITY\SYSTEM</Data></EventData></Event> host = WIN11-TARGET source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational
>	6/22/26 2:06:31.407 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698f7bd9}' /><EventID>13</EventID><Version>2</Version><Level>4</Level><Task>13</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-06-22T18:06:31.4072664Z' /><EventRecordID>34930</EventRecordID><Correlation><Execution ProcessID='3176' ThreadID='4132' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>WIN11-TARGET</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='RuleName'>technique_id=T1053, technique_name=Scheduled Task</Data><Data Name='EventType'>SetValue</Data><Data Name='UtcTime'>2026-06-22 18:06:31.408</Data><Data Name='ProcessGuid'>{878543f7-7907-6a39-1b00-000000000000}</Data><Data Name='ProcessId'>1396</Data><Data Name='Image'>C:\WINDOWS\system32\svchost.exe</Data><Data Name='TargetObject'>HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tree\WindowsUpdateCheck\id</Data><Data Name='Details'>{C66421A4-5553-4D4A-A8ED-77A795FC800F}</Data><Data Name='User'>NT AUTHORITY\SYSTEM</Data></EventData></Event> host = WIN11-TARGET source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational
>	6/22/26 2:06:31.407 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698f7bd9}' /><EventID>13</EventID><Version>2</Version><Level>4</Level><Task>13</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-06-22T18:06:31.4071893Z' /><EventRecordID>34929</EventRecordID><Correlation><Execution ProcessID='3176' ThreadID='4132' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>WIN11-TARGET</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='RuleName'>technique_id=T1053, technique_name=Scheduled Task</Data><Data Name='EventType'>SetValue</Data><Data Name='UtcTime'>2026-06-22 18:06:31.408</Data><Data Name='ProcessGuid'>{878543f7-7907-6a39-1b00-000000000000}</Data><Data Name='ProcessId'>1396</Data><Data Name='Image'>C:\WINDOWS\system32\svchost.exe</Data><Data Name='TargetObject'>HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tree\WindowsUpdateCheck\SD</Data><Data Name='Details'>Binary Data</Data><Data Name='User'>NT AUTHORITY\SYSTEM</Data></EventData></Event> host = WIN11-TARGET source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational

Detection screenshot: *Figure 3: Splunk detection of T1053.005 — events showing schtasks.exe spawned by PowerShell creating WindowsUpdateCheck persistence task*

Searches, Reports, and Alerts										
Searches, reports, and alerts are saved searches created from pivot or the search page. Learn more New Report New Alert										
1 Searches, Reports, and Alerts Type: All App: Search & Reporting (search) Owner: Administrator (admin) T1053 10 per page										
Name	Actions	Type	Next Scheduled Time	Display View	Owner	App	Alerts	Sharing	Status	
Suspicious Scheduled Task Creation - T1053.005 Detects scheduled task creation via schtasks.exe spawned by PowerShell. MITRE ATT&CK T1053.005.	Edit Run View Recent	Alert	2026-06-23 19:00:00 Eastern Daylight Time	none	admin	search	0	Private	Enabled	

Alert screenshot: *Figure 4: Splunk alert rule "Suspicious Scheduled Task Creation - T1053.005" configured to run hourly*

Recommended Mitigation

- Monitor scheduled task creation events (Windows Event ID 4698) in production
- Alert on any scheduled task created to run as SYSTEM by a non-administrative process
- Regularly audit scheduled tasks using schtasks /query to identify unauthorized entries
- Restrict task creation permissions for standard user accounts via Group Policy

Technique 3: Masquerading — T1036.005

Field	Details
Technique	Masquerade Task or Service
MITRE ID	T1036.005
Tactic	Defense Evasion
Severity	High

Description

Masquerading is a defense evasion technique where attackers rename malicious files to mimic legitimate Windows processes, making it harder for analysts and security tools to identify malicious activity. A common approach is naming a malicious executable svchost.exe — one of the most frequently running Windows processes — and executing it from a non-standard location.

Attack Simulation

The following commands were executed to simulate masquerading on WIN11-TARGET:

```
Copy-Item "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -Destination "C:\Users\analyst\AppData\Local\Temp\svchost.exe"
```

```
Start-Process "C:\Users\analyst\AppData\Local\Temp\svchost.exe" -ArgumentList "-WindowStyle Hidden -Command whoami"
```

PowerShell was copied to the Temp folder and renamed svchost.exe, then executed from that non-standard location — mimicking how real attackers disguise malicious payloads.

Detection

The key detection indicator is **path anomaly** — legitimate svchost.exe exclusively runs from C:\Windows\System32\. Any instance running from another location is immediately suspicious. Sysmon captured 179 events including:

Sysmon Event	ID	What It Captured
File Creation	EventID 11	svchost.exe written to Temp folder
Process Creation	EventID 1	svchost.exe executing from Temp
Process Termination	EventID 5	Fake svchost process ended
Process Access	EventID 10	DLL injection attempt detected (T1055.001)

The following Splunk query was used to detect the activity:

index=main source="WinEventLog:Microsoft-Windows-Sysmon/Operational" "svchost.exe" "Temp"

i	Time	Event
>	6/23/26 6:00:01.603 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698ffbdc}' /><EventID>11</EventID><Version>2</Version><Level>4</Level><Task>11</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-06-23T22:00:01.603633Z' /><EventRecordID>40661</EventRecordID><Correlation><Execution ProcessID='3452' ThreadID='2408' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>WIN11-TARGET</Computer><Security UserID='S-1-5-18' /><System><EventData><Data Name='RuleName'>technique_id=T1574.010, technique_name=Services File Permissions Weakness</Data><Data Name='UtcTime'>2026-06-23 21:59:54.248</Data><Data Name='ProcessGuid'>{878543f7-771c-6a3a-2801-000000000000}</Data><Data Name='ProcessId'>3760</Data><Data Name='Image'>C:\WINDOWS\system32\svchost.exe</Data><Data Name='TargetFileName'>C:\Windows\Temp\pool_tags_summary.js on.dup</Data><Data Name='CreationUtcTime'>2026-06-23 21:59:54.238</Data><Data Name='User'>NT AUTHORITY\SYSTEM</Data></EventData></Event> host = WIN11-TARGET source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational
>	6/23/26 5:59:57.852 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698ffbdc}' /><EventID>11</EventID><Version>2</Version><Level>4</Level><Task>11</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-06-23T21:59:57.8525801Z' /><EventRecordID>40651</EventRecordID><Correlation><Execution ProcessID='3452' ThreadID='2408' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>WIN11-TARGET</Computer><Security UserID='S-1-5-18' /><System><EventData><Data Name='RuleName'>technique_id=T1574.010, technique_name=Services File Permissions Weakness</Data><Data Name='UtcTime'>2026-06-23 21:59:50.106</Data><Data Name='ProcessGuid'>{878543f7-771c-6a3a-2801-000000000000}</Data><Data Name='ProcessId'>3760</Data><Data Name='Image'>C:\WINDOWS\system32\svchost.exe</Data><Data Name='TargetFileName'>C:\Windows\Temp\pool_tags_summary.js on</Data><Data Name='CreationUtcTime'>2026-06-11 00:13:14.978</Data><Data Name='User'>NT AUTHORITY\SYSTEM</Data></EventData></Event> host = WIN11-TARGET source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational
>	6/23/26 5:25:20.838 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698ffbdc}' /><EventID>11</EventID><Version>2</Version><Level>4</Level><Task>11</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-06-23T21:25:20.8384930Z' /><EventRecordID>40175</EventRecordID><Correlation><Execution ProcessID='3452' ThreadID='2408' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>WIN11-TARGET</Computer><Security UserID='S-1-5-18' /><System><EventData><Data Name='RuleName'>technique_id=T1574.010, technique_name=Services File Permissions Weakness</Data><Data Name='UtcTime'>2026-06-23 21:25:20.832</Data><Data Name='ProcessGuid'>{878543f7-f693-6a3a-4f00-000000000000}</Data><Data Name='ProcessId'>3792</Data><Data Name='Image'>C:\WINDOWS\system32\svchost.exe</Data><Data Name='TargetFileName'>C:\Windows\Temp\APPX.8161901wx5g5wki8uzcpkr4.tmp</Data><Data Name='CreationUtcTime'>2026-06-23 21:25:20.832</Data><Data Name='User'>NT AUTHORITY\SYSTEM</Data></EventData></Event> host = WIN11-TARGET source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational
>	6/23/26 5:25:20.706 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698ffbdc}' /><EventID>11</EventID><Version>2</Version><Level>4</Level><Task>11</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-06-23T21:25:20.7069721Z' /><EventRecordID>40174</EventRecordID><Correlation><Execution ProcessID='3452' ThreadID='2408' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>WIN11-TARGET</Computer><Security UserID='S-1-5-18' /><System><EventData><Data Name='RuleName'>technique_id=T1574.010, technique_name=Services File Permissions Weakness</Data><Data Name='UtcTime'>2026-06-23 21:25:20.697</Data><Data Name='ProcessGuid'>{878543f7-f693-6a3a-4f00-000000000000}</Data><Data Name='ProcessId'>3792</Data><Data Name='Image'>C:\WINDOWS\system32\svchost.exe</Data><Data Name='TargetFileName'>C:\Windows\Temp\APPX.vkad4f6myhwxiwg46d_lbh.tmp</Data><Data Name='CreationUtcTime'>2026-06-23 21:25:20.697</Data><Data Name='User'>NT AUTHORITY\SYSTEM</Data></EventData></Event> host = WIN11-TARGET source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational

Detection screenshot: Figure 5: Splunk detection of T1036.005 — 179 events showing svchost.exe executing from Temp folder, mapped to T1036.005 and T1055.001

Searches, Reports, and Alerts									
Searches, reports, and alerts are saved searches created from pivot or the search page. Learn more									
1 Searches, Reports, and Alerts Type: All App: Search & Reporting Owner: Administrator (admin) T1036.005 10 per page									
Name	Actions	Type	Next Scheduled Time	Display View	Owner	App	Alerts	Sharing	Status
Masquerading Detection - T1036.005 Detects svchost.exe running from non-standard location - MITRE ATT&CK T1036.005	Edit Run View Recent	Alert	2026-06-23 19:00:00 Eastern Daylight Time	none	admin	search	3	Private	✓ Enabled

Alert screenshot: Figure 6: Splunk alert rule "Masquerading Detection - T1036.005" configured to run hourly

Notable Finding: Sysmon also automatically detected a secondary technique — T1055.001 (Dynamic-link Library Injection) — triggered by the masquerading process attempting to access legitimate Windows processes. This demonstrates how a single attacker action can generate detections across multiple MITRE techniques simultaneously.

Recommended Mitigation

- Implement process path validation rules — alert on any known Windows system process running outside its expected directory
- Deploy application whitelisting to prevent execution of unauthorized binaries from user-writable locations such as Temp folders

- Block script execution from AppData and Temp directories via Software Restriction Policies or AppLocker
- Monitor for process names that match known Windows binaries but run from unexpected paths

Conclusion

This simulation demonstrates end-to-end detection capability across three MITRE ATT&CK tactics using an isolated Windows 11 lab environment. All three techniques were successfully simulated, detected via Sysmon logs ingested into Splunk, and mapped to their corresponding MITRE ATT&CK technique IDs. Scheduled detection rules were deployed to provide ongoing alerting capability.

The exercise highlights several key findings relevant to real-world security operations:

A single attacker command can trigger detections across multiple MITRE techniques simultaneously, as demonstrated by the masquerading simulation which also surfaced a DLL injection detection. Sysmon with a professionally tuned configuration provides automatic MITRE ATT&CK tagging, significantly accelerating analyst triage. The parent-child process relationship visible in Sysmon logs is a critical indicator — legitimate Windows processes have predictable parent processes, and anomalies in these relationships are a reliable signal of malicious activity.

Defensive Recommendations Summary

Technique	Priority	Key Control
T1059.001 PowerShell	High	Enforce execution policy, enable script block logging
T1053.005 Scheduled Task	High	Monitor task creation events, audit existing tasks
T1036.005 Masquerading	High	Process path validation, application whitelisting

The screenshot shows the Splunk 'Searches, Reports, and Alerts' page. It displays a list of three scheduled alerts, each corresponding to a MITRE ATT&CK technique. The alerts are: 'Masquerading Detection - T1036.005', 'Suspicious PowerShell Execution - T1059.001', and 'Suspicious Scheduled Task Creation - T1053.005'. Each alert is configured to run on 2025-06-22 at 17:00:00 Eastern Daylight Time, with a display view of 'None', owner 'admin', app 'search', and 1 alert. All alerts are in 'Enabled' status.

Name	Actions	Type	Next Scheduled Time	Display View	Owner	App	Alerts	Sharing	Status
Masquerading Detection - T1036.005 Detects suspicious user running from non-standard location - MITRE ATT&CK T1036.005	Edit Run View Report	Alert	2025-06-22 17:00:00 Eastern Daylight Time	None	admin	search	1	Private	✓ Enabled
Suspicious PowerShell Execution - T1059.001 Detects PowerShell execution with &commandPolicy, System log - MITRE ATT&CK T1059.001	Edit Run View Report	Alert	2025-06-22 17:15:00 Eastern Daylight Time	None	admin	search	0	Private	✓ Enabled
Suspicious Scheduled Task Creation - T1053.005 Detects scheduled task creation via cmd.exe and spawned by PowerShell - MITRE ATT&CK T1053.005	Edit Run View Report	Alert	2025-06-22 17:00:00 Eastern Daylight Time	None	admin	search	1	Private	✓ Enabled

Figure 7: All three MITRE ATT&CK detection rules active in Splunk — T1059.001, T1053.005, and T1036.005